

HW#9

REF: Read Theory Manual Chapter 2: §2.7 Reactivity Coefficients through §2.9 Components of Net Reactivity; (pages 10 through 22)
Module #9 PPT Slides

1. Which ONE of the following best describes what happens to the Fast Non-Leakage Probability ($\mathcal{L}_f$) if moderator temperature is decreased?
 - a. Increases due to the longer neutron slowing down length.
 - b. Decreases due to the longer neutron slowing down length.
 - c. Increases due to the shorter neutron slowing down length.
 - d. Decreases due to the shorter neutron slowing down length.
2. If moderator temperature is increased, which of the following best describes what happens to the Resonance Escape Probability (p)?
 - a. Increases due to spectrum hardening resulting in increased resonance absorption.
 - b. Decreases due to spectrum hardening resulting in increased resonance absorption.
 - c. Increases due to a decrease in the macroscopic scattering cross section Σ_s of the moderator.
 - d. Increases due to a decrease in the macroscopic absorption cross section Σ_a of the moderator.
3. Which ONE of the following best describes what happens to Thermal Utilization (f) if moderator temperature is increased?
 - a. Increases due to spectrum hardening resulting in increased resonance absorption.
 - b. Decreases due to spectrum hardening resulting in increased resonance absorption.
 - c. Increases due to a decrease in the macroscopic scattering cross section Σ_s of the moderator.
 - d. Increases due to a decrease in the macroscopic absorption cross section Σ_a of the moderator.
4. Suppose the reactor is sub-critical with $K_{eff} = 0.9999$ with reactor pool (moderator) temp at 100 °F. Assume that the temperature coefficient of reactivity (α_{Tmod}) = -3.9 pcm / °F. Calculate the resultant reactor period (τ) if the pool temperature cooled down from 100°F to 91°F.

-
5. The void coefficient (α_{voids}) = -1.6 pcm/cc for the PULSTAR. Suppose the PULSTAR is operating and is critical at 1 MW, and a fuel cooling channel suddenly becomes blocked causing the water in the channel to boil. Which of the following scenarios would occur?
 - a) The PULSTAR is slightly over-moderated, so boiling would cause a negative reactivity effect causing reactor power to decrease.
 - b) The PULSTAR is slightly over-moderated, so boiling would cause a positive reactivity effect causing reactor power to increase.
 - c) The PULSTAR is slightly under-moderated, so boiling would cause a negative reactivity effect causing reactor power to decrease.
 - d) The PULSTAR is slightly under-moderated, so boiling would cause a positive reactivity effect causing reactor power to increase.

 6. When the PULSTAR reactor power increases from 10kW to 900kW:
 - a) Doppler broadening of absorption resonances in ^{235}U decreases causing K_{eff} to increase.
 - b) Doppler broadening of absorption resonances in ^{238}U increases adding negative reactivity.
 - c) Doppler broadening of absorption resonances in ^{235}U increases causing the resonance escape probability (p) to increase.
 - d) Doppler broadening of absorption resonances in ^{238}U decreases adding positive reactivity.

 7. As reactor fuel temperature increases:
 - a) The resonances of ^{238}U widen and the area under each resonance increases.
 - b) The resonances of ^{238}U widen and the area under each resonance stays the same.
 - c) The resonances of ^{238}U widen and the area under each resonance decreases.
 - d) The resonances of ^{238}U widen causing thermal utilization (f) to increase.

 8. Xenon fill in the blanks:
 - a) The thermal neutron absorption cross section of ^{135}Xe is _____ barns.
 - b) Following a change in power level, it takes ~_____ hours (~ 5 half lives) for the ^{135}Xe concentration to reach equilibrium.
 - c) Following a change in power level, it takes ~_____ hours (~ 5 half lives) for the ^{135}I concentration to reach equilibrium.
 - d) Following a reactor shutdown, it takes ~_____ hours for the ^{135}Xe concentration to decay away to zero.
 - e) If reactor power is held stable at 1 MW for 50 hours, and is then decreased to 500 kW, burnup of ^{135}Xe immediately _____ causing the ^{135}Xe concentration to _____ due to the decay of _____.

9. **Do Chapt #2 Prob #9 on page 32** (*use a separate sheet of paper*).